**TATA POWER**
Lighting Up Lives!

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# EV CHARGING
# ANALYTICS DASHBOARD

Data Analysis & Insights Presentation


Prepared By

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DATA ANALYSIS | INSIGHTS | EV NETWORK | PERFORMANCE

Powering a Greener Tomorrow

PROJECT INTRODUCTION

TATA POWER

# TATA POWER EV CHARGING
## PERFORMANCE ANALYSIS DASHBOARD

Data-driven insights for smarter expansion and higher profitability

### PROJECT OVERVIEW
This project analyzes the performance and expansion potential of Tata Power EV charging infrastructure **across multiple districts.**

### DASHBOARD PURPOSE
The dashboard combines operational, customer usage, and financial ROI metrics to support **data-driven business decisions.**

### OBJECTIVE
To identify high-demand locations, optimize station utilization, and **improve profitability**.

### ANALYSIS FOCUS
The analysis focuses on **charging demand trends**, customer behavior, operational efficiency, and financial performance.

### BUSINESS IMPACT
Insights from this dashboard can help Tata Power plan future EV charging station **expansion strategically**.

### TOOLS & TECHNIQUES
The project was developed using **Excel Dashboards, KPI Analysis**, Pivot Tables, Slicers, and Data Visualization Techniques.


| PERFORMANCE | CUSTOMER | OPERATIONAL | FINANCIAL |
|---|---|---|---|
| Monitoring | Insights | Excellence | ROI |

# DATA METHODOLOGY

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**Business and Usage Overview**

**Dashboard 1**

**Station Utilization and Operational Efficiency**

**Dashboard 2**

**Customer and Demand Behaviour**

**Dashboard 3**

**Financial Performance and Expansion Strategy (ROI Focus)**

**Dashboard 4**

# BUSINESS AND USAGE OVERVIEW

## PROBLEMS TO ADDRESS

- How is charging demand evolving over time?
- Are stations able to handle growing demand?
- What is the charging behavior pattern by time of the day?

## IMPORTANT KPIs

- Total charging sessions
- Total energy consumed (kWh charged)
- Total revenue (from charging sessions)
- Number of active customers
- Average revenue per session
- Average kWh per session

# KEY INSIGHTS OVERVIEW

Demand Trends, Station Capacity & Usage Behavior

## 1 How is charging demand evolving over time?

### Key Insights

- EV charging demand showed steady growth from Aug to Oct 2024.
- Average daily demand remained stable at ~31K–33K kWh.
- October had more high-demand days, reflecting rising EV adoption.
- Weekday demand is consistently higher than weekends.

### Important Findings

- Highest demand: Wed (~422K kWh) & Tue (~418K kWh)
- Lowest demand: Sat (~383K kWh) & Fri (~385K kWh)

### Business Conclusion

- Public EV charging dependency is increasing.
- Demand growth is becoming more stable and consistent.
- Urban & commercial charging is the key driver of growth.

## 2 Are stations able to handle growing demand?

### Key Insights

- Current charging stations are handling demand efficiently, but high utilization in some urban locations.

**High Demand Locations**

| Area | Current Plugs | Projected Plugs |
|---|---|---|
| Jayanagar | 30 | 109 |
| Rajajinagar | 30 | 103 |
| Indiranagar | 30 | 97 |
| HSR Layout | 24 | 94 |
| Koramangala | 24 | 88 |

### Important Findings

- Commercial & IT hubs are driving the highest charging activity.
- Low-demand areas have sufficient charging capacity.

### Business Conclusion

- Existing infrastructure is stable for current operations.
- Rapid expansion of stations & plugs is needed in high-demand urban areas.

## 3 What is the charging behavior pattern by time of the day?

### Key Insights

- Charging demand peaks between 08 AM – 03 PM with highest activity 10 AM – 03 PM.
- Overnight charging demand remains significantly lower.
- Weekday utilization is stronger than weekends.

**Hourly Charging Trend**

| Time Period | Charging Trend |
|---|---|
| 12 AM – 07 AM | Low demand |
| 08 AM – 03 PM | Peak demand |
| 04 PM – 07 PM | Moderate demand |
| After 08 PM | Demand declines |

### Important Findings

- Most users prefer charging during office & travel hours.
- Workplace & commercial-area charging contributes the highest usage.

### Business Conclusion

- Optimizing daytime charging infrastructure is essential.
- Fast chargers should be prioritized in commercial & business hubs.
- Off-peak pricing strategies can help shift demand toward nighttime hours.

# KPIs MEASURES

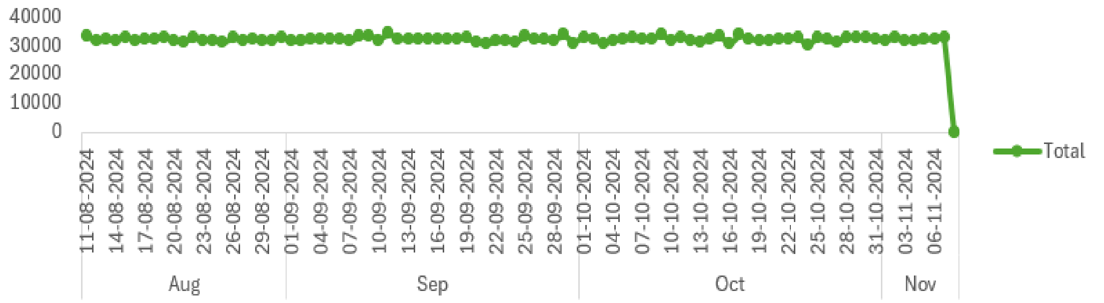
| | | | |
|---|---|---|---|
| 1 | Total Charging Session | = | COUNT('Charging Session'[station_id]) |
| 2 | Total energy consumed (kWh charged) | = | SUM('Charging Session'[kwh_charged]) |
| 3 | Total revenue (from charging sessions) | = | SUM('Charging Session'[total_cost]) |
| 4 | Number of active customers | = | COUNT(Customer[customer_id]) |
| 5 | Average revenue per session | = | [Total revenue (from charging sessions)] / COUNT('Charging Session'[session_id]) |
| 6 | Average kWh per session | = | SUM('Charging Session'[kwh_charged]) / COUNT('Charging Session'[session_id]) |

# Business and Usage Overview

| | |
|---|---|
| 1,14,000 | Total Charging Sessions |
| 28,59,771.70 | Total Revenue (in rs) |
| 4,96,93,947 | Average Revenue Per Session (in rs) |
| 8,000 | Total Energy Consumed (in Kwh) |
| 435.91 | Total Active Customer |
| 25.09 | Average Kwh Per Session |

Sum of kwh_charged

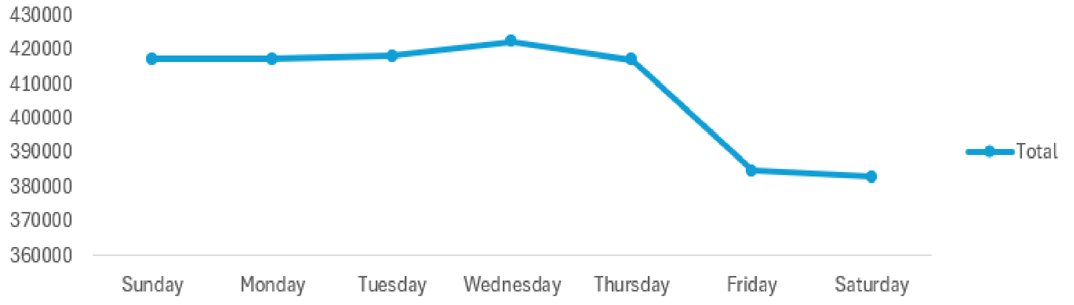
## Daily Energy Consumption Trend (kWh)




session_start_Date (Month) ▼ session_start_Date ▼

Sum of kwh_charged

## Energy Usage by Day of Week



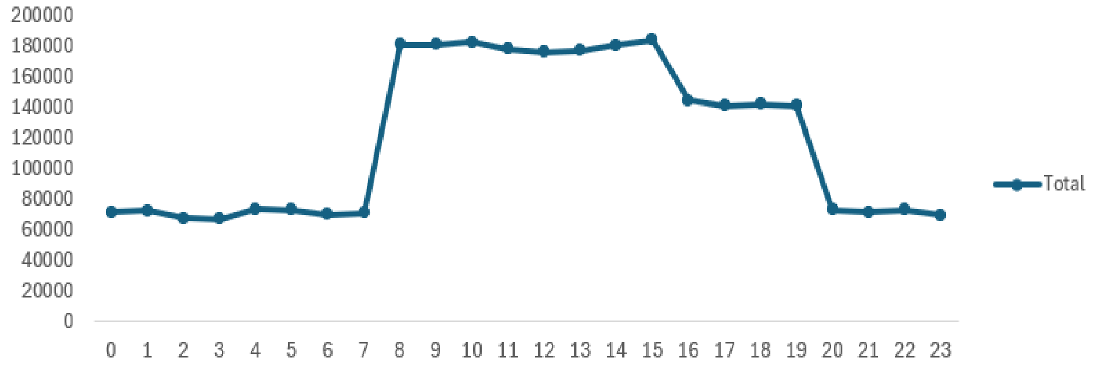

Day Name ▼

Day Name

- Wednesday
- Tuesday
- Thursday
- Sunday
- Saturday
- Monday
- Friday

Sum of kwh_charged

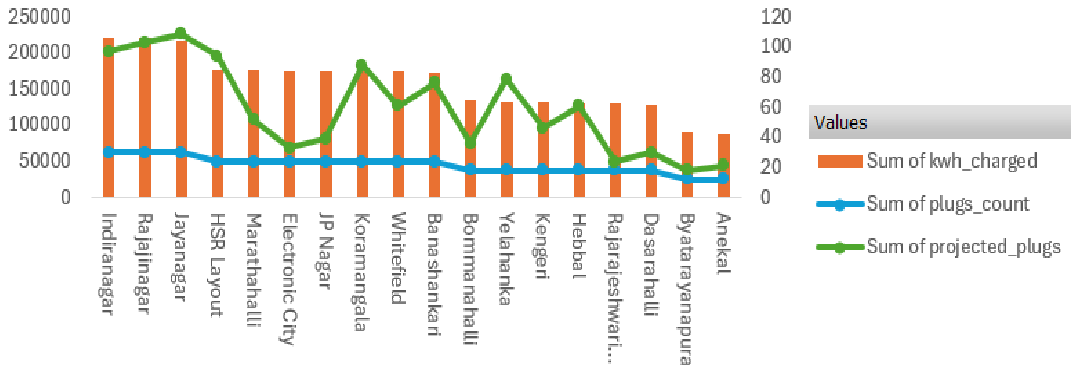
## Hourly Charging Demand Pattern




Charging Hour ▼

Sum of kwh_charged | Sum of plugs_count | Sum of projected_plugs

## District-wise Demand vs Charging Infrastructure




district_name ▼

district_n...

- Anekal
- Banashank...
- Bommana...
- Byatarayan...
- Dasarahalli
- Electronic ...
- Hebbal
- HSR Layout

# STATION UTILIZATION AND OPERATIONAL EFFICIENCY

## PROBLEMS TO ADDRESS

- Which stations are overutilized and may require capacity expansion
- Which stations are underutilized and need operational improvement
- Are available plugs being efficiently used across locations
- What is the load distribution across districts and stations

## IMPORTANT KPIs

- Number of sessions per station
- Average utilization per station (sessions per plug)
- Plugs count vs actual usage
- Average kWh per station
- District wise station density
- Sessions per plug ratio

# EV CHARGING STATION ANALYSIS

Key Insights for Infrastructure Planning & Operational Optimization

## 1 WHICH STATIONS ARE OVERUTILIZED AND MAY REQUIRE CAPACITY EXPANSION?

### KEY INSIGHTS

- Indiranagar, Rajajinagar, and Jayanagar show the highest charging demand and utilization.
- Projected plug requirements are significantly higher than current infrastructure, indicating future capacity pressure.
- Commercial and high-density urban areas are experiencing the fastest demand growth.

### HIGH-DEMAND STATIONS

- **Indiranagar** (~223K kWh)
- **Rajajinagar** (~221K kWh)
- **Jayanagar** (~217K kWh)
- **HSR Layout & Koramangala** Strong utilization growth

### BUSINESS CONCLUSION

High-demand urban hubs will require rapid charging infrastructure expansion to avoid congestion and waiting-time issues.

## 2 WHICH STATIONS ARE UNDERUTILIZED AND NEED OPERATIONAL IMPROVEMENT?

### KEY INSIGHTS

- Stations such as Anekal, Byatarayanapura, and Dasarahalli show lower charging utilization.
- Demand growth in these locations is slower compared to major urban hubs.
- Plug utilization efficiency is relatively low in suburban and lower-traffic areas.

### IMPROVEMENT SUGGESTIONS

- Increase customer awareness and accessibility
- Improve charger visibility and navigation support
- Introduce promotional pricing and loyalty programs

### BUSINESS CONCLUSION

Low-demand locations currently require operational optimization rather than immediate infrastructure expansion.

# EV CHARGING STATION ANALYSIS

Key Insights for Efficient Utilization & Load Distribution

## 3 ARE AVAILABLE PLUGS BEING EFFICIENTLY USED ACROSS LOCATIONS?

### KEY INSIGHTS

- Session distribution across major stations is relatively balanced.
- Stations with similar plug counts are handling similar charging volumes.
- No major imbalance in plug utilization is observed.

### HIGHEST SESSION UTILIZATION STATIONS

Station IDs: **24, 7, 25, 23,** and **21** recorded the highest session activity.


### BUSINESS CONCLUSION

Current plug infrastructure is being utilized efficiently, showing good load balancing and stable operational performance. However, high-demand stations may require additional plugs in the future.

## 4 WHAT IS THE LOAD DISTRIBUTION ACROSS DISTRICTS AND STATIONS?

### KEY INSIGHTS

- Highest charging load is concentrated in Indiranagar, Rajajinagar, Jayanagar, HSR Layout, and Marathahalli.
- Commercial hubs, IT corridors, and dense residential areas contribute the highest demand.
- Peripheral districts such as Anekal and Byatarayanapura show lower utilization.

### HIGHEST LOAD DISTRICTS

| | |
|---|---|
| Indiranagar | ~222K |
| Rajajinagar | ~221K |
| Jayanagar | ~218K |
| HSR Layout | ~178K |
| Marathahalli | ~178K |

### MODERATE LOAD DISTRICTS

| | |
|---|---|
| Whitefield | ~175K |
| Koramangala | ~175K |
| JP Nagar | ~175K |
| Electronic City | ~176K |
| Banashankari | ~174K |

### LOWER LOAD DISTRICTS

| | |
|---|---|
| Anekal | ~88K |
| Byatarayanapura | ~90K |
| Dasarahalli | ~129K |

### BUSINESS CONCLUSION

Charging demand is unevenly distributed across districts. Infrastructure expansion should focus on high-load urban zones, while low-load areas should prioritize awareness and operational improvements.

# KPIs MEASURES

| # | KPI | | Numerator | Denominator |
|---|---|---|---|---|
| 1 | Number of sessions per station | = | COUNT('Charging Session'[session_id]) | COUNT('Charging Station'[station_id]) |
| 2 | Average utilization per station (sessions per plug) | = | COUNT('Charging Session'[session_id]) | SUM('Charging Station'[plugs_count]) |
| 3 | Plugs count vs actual usage | = | SUM('Charging Session'[kwh_charged]) | SUM('Charging Station'[plugs_count]) |
| 4 | Average kWh per station | = | SUM('Charging Session'[kwh_charged]) | COUNT('Charging Station'[station_id]) |
| 5 | District wise station density | = | COUNT('Charging Station'[station_id]) | COUNT(Districts[district_name]) |

# Station Utilization and Operational Efficiency

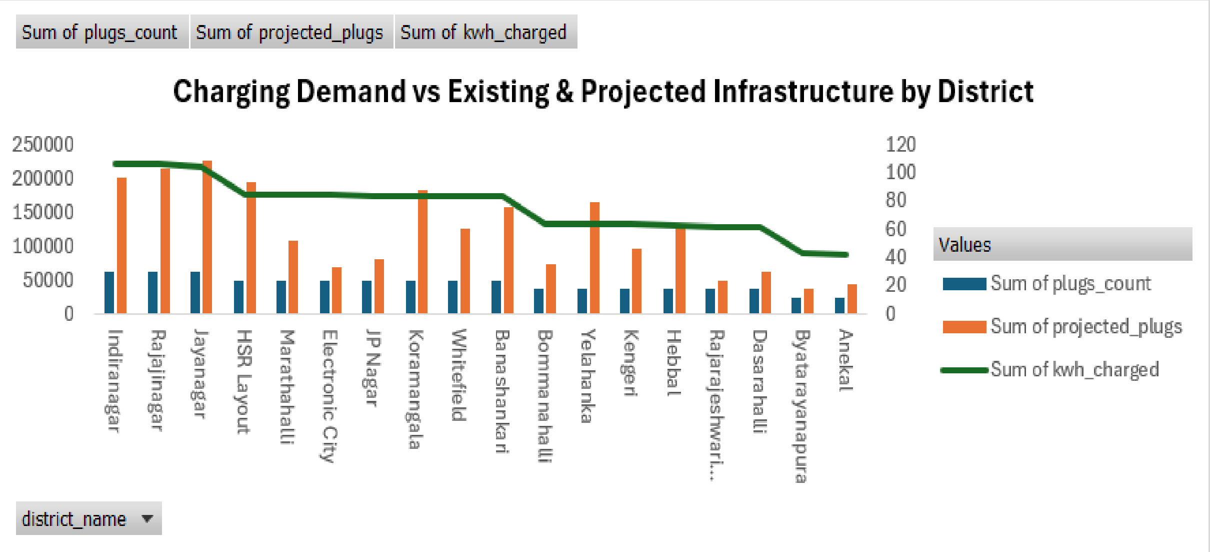
**1,754**
Session Per Station

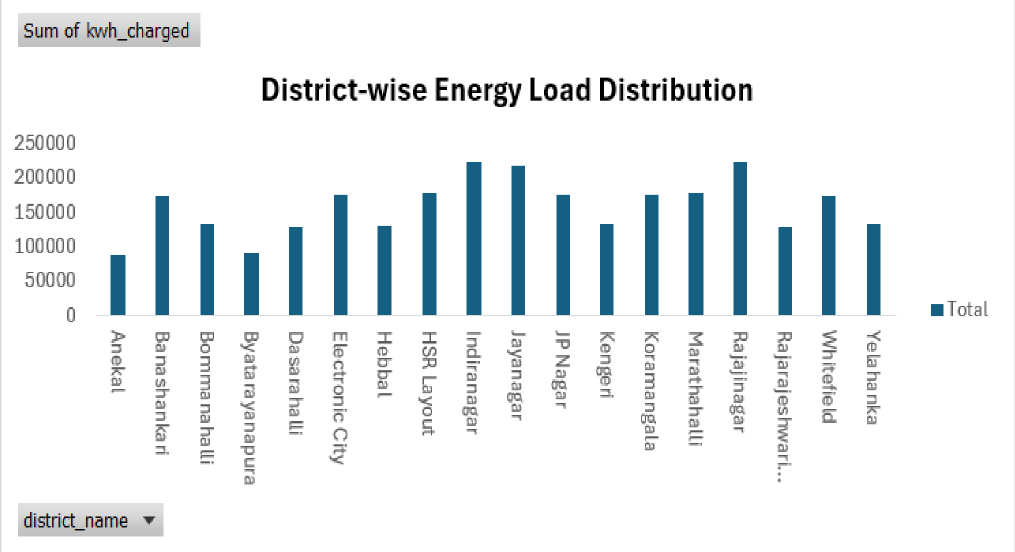
**292**
Session Per Plug

**7,333**
Plugs Count Vs Actual Usage

**43,996**
Average kwh per station (In Kwh)

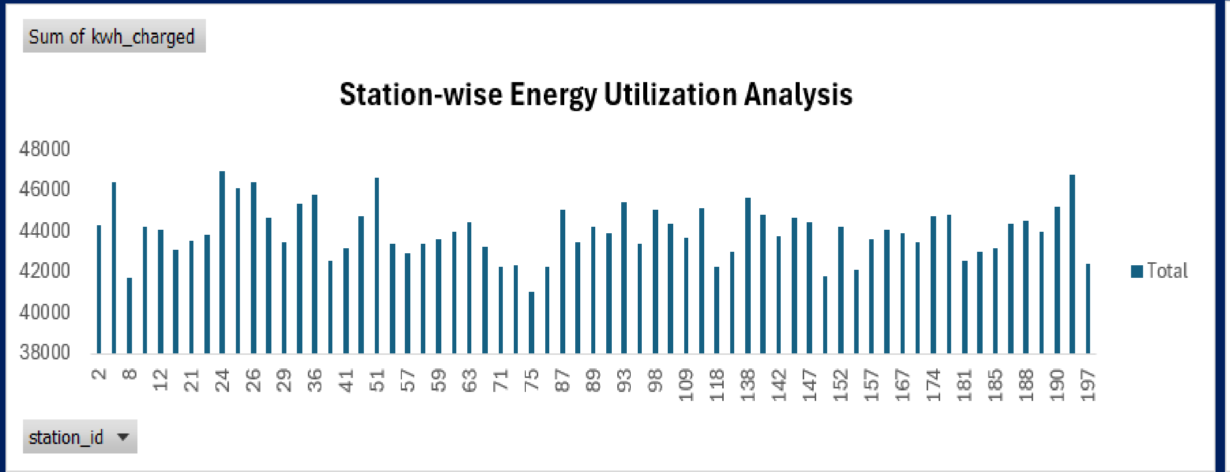
**4**
District wise station density

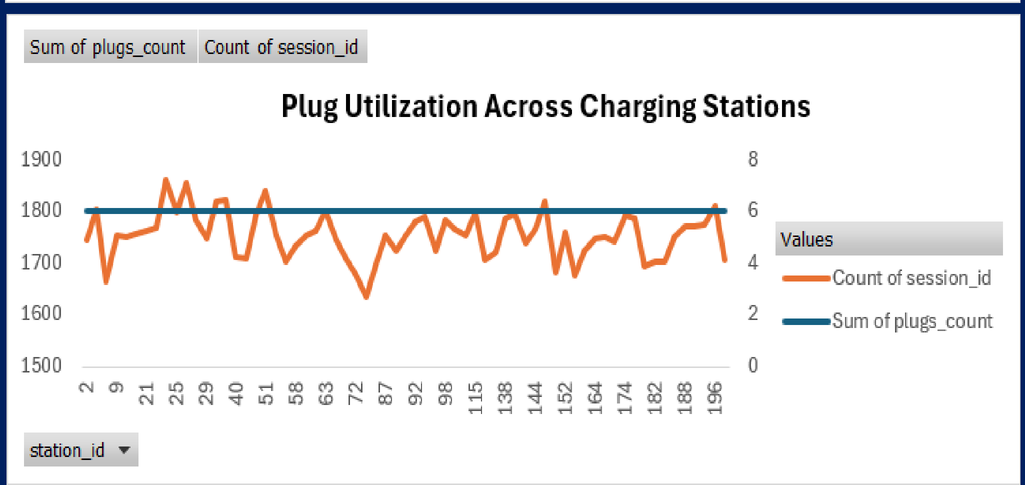



district_name

- Banashankari
- Bommanahalli
- Byatarayanapura
- Dasarahalli
- Electronic City
- Hebbal
- HSR Layout
- Indiranagar




station_id

- 2
- 7
- 8
- 9
- 12
- 18
- 21

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# CUSTOMER AND DEMAND BEHAVIOR

## PROBLEMS TO ADDRESS

- What type of customers contribute most to usage and revenue?
- How does income tier influence charging behavior?
- What are the usage patterns across different EV models and battery capacities?
- Are there segments with high potential but low engagement?

## KPIs TO TRACK

- Unique customers count
- Sessions per customer
- Average kWh per customer
- Revenue per customer
- Usage by income tier
- Usage by car model and battery capacity

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# CUSTOMER & BEHAVIOR INSIGHTS

Understanding Customer Contribution and Charging Behavior

## 1 WHAT TYPE OF CUSTOMERS CONTRIBUTE MOST TO USAGE AND REVENUE?

### KEY INSIGHTS

- Mid-Range customers are the highest contributors in charging usage and revenue.
- Low-Mid income users are the second-largest contributor to public charging demand.
- High-income customers show comparatively lower dependency on public charging infrastructure.

### BUSINESS INTERPRETATION

Mid-Range and Low-Mid users mainly represent daily commuters and frequent public charging users.

High-income customers are more likely to rely on home or private charging solutions.

### BUSINESS CONCLUSION

Mid-Range and Low-Mid income groups are the primary drivers of EV charging demand and revenue generation.

## 2 HOW DOES INCOME TIER INFLUENCE CHARGING BEHAVIOR?

### KEY INSIGHTS

- Mid-Range users show the highest charging frequency and public station utilization.
- Low-Mid customers also contribute significantly, reflecting increasing EV accessibility.
- High-income users utilize public charging less frequently despite higher purchasing power.

### BEHAVIORAL PATTERN

**MID-RANGE USERS**

- Mainly charge during work and travel hours.
- High reliance on public charging infrastructure.

**HIGH-INCOME CUSTOMERS**

- Prefer private/home charging.
- Use premium infrastructure selectively.

### BUSINESS CONCLUSION

Income tier strongly impacts charging frequency, public charging dependency, and revenue contribution.

# CUSTOMER & USAGE PATTERNS

Insights on EV Models, Battery Capacity & Growth Opportunities

## 3 WHAT ARE THE USAGE PATTERNS ACROSS DIFFERENT EV MODELS AND BATTERY CAPACITIES?

**HIGHEST CHARGING DEMAND EV MODELS**

| EV MODEL | kWh CHARGED | AVG BATTERY CAPACITY |
|---|---|---|
| Mahindra XUV400 EC Pro | ~158K | 39.4 kWh |
| Hyundai Kona Electric | ~157K | 39.2 kWh |
| Tata Punch EV | ~145K | 35 kWh |
| Citroen eC3 | ~144K | 29.2 kWh |
| Ampere Nexus | ~138K | 30 kWh |
| Tata Nexon EV | ~136K | 30.2 kWh |

**PREMIUM / LARGE BATTERY EVs**

| EV MODEL | BATTERY CAPACITY |
|---|---|
| Audi e-tron 55 | 95 kWh |
| Kia EV6 | 77.4 kWh |
| Volvo XC40 Recharge | 75 kWh |
| Hyundai Ioniq 5 | 72.6 kWh |

**OBSERVATION**

- Larger battery EVs consume higher energy per session.
- Mid-range battery EVs dominate total charging frequency.

**KEY INSIGHTS**

- Popular urban EV models contribute the highest charging activity.
- Vehicles with moderate battery capacity (25–40 kWh) and daily commuting usage dominate station utilization.
- Premium high-battery EVs contribute strong energy consumption but comparatively lower overall charging frequency.

**BUSINESS CONCLUSION**

Mid-capacity commuter EVs are the major contributors to charging demand, while premium EVs contribute higher energy usage per session.

## 4 ARE THERE SEGMENTS WITH HIGH POTENTIAL BUT LOW ENGAGEMENT?

**KEY INSIGHTS**

- High-income customers and premium EV owners show lower public charging usage despite strong purchasing power.
- Locations such as Anekal and Byatarayanapura have lower station utilization.
- Night-time charging demand remains comparatively low, leaving unused charging capacity.

**GROWTH OPPORTUNITIES**

- Introduce premium charging plans and off-peak discounts.
- Improve awareness campaigns and customer education.
- Invest in faster and smarter charging infrastructure.

**BUSINESS CONCLUSION**

High-income users, premium EV owners, and off-peak charging segments offer strong future growth opportunities.

# KPIs MEASURES

1. **Unique customers count** = COUNT(Customer[customer_id])

2. **Sessions per customer** = $$\frac{\text{COUNT('Charging Session'[session\_id])}}{\text{COUNT(Customer[customer\_id])}}$$

3. **Average kWh per customer** = $$\frac{\text{[Sum of kwh\_charged]}}{\text{COUNT(Customer[customer\_id])}}$$

4. **Revenue per customer** = $$\frac{\text{[Sum of total\_cost]}}{\text{COUNT(Customer[customer\_id])}}$$

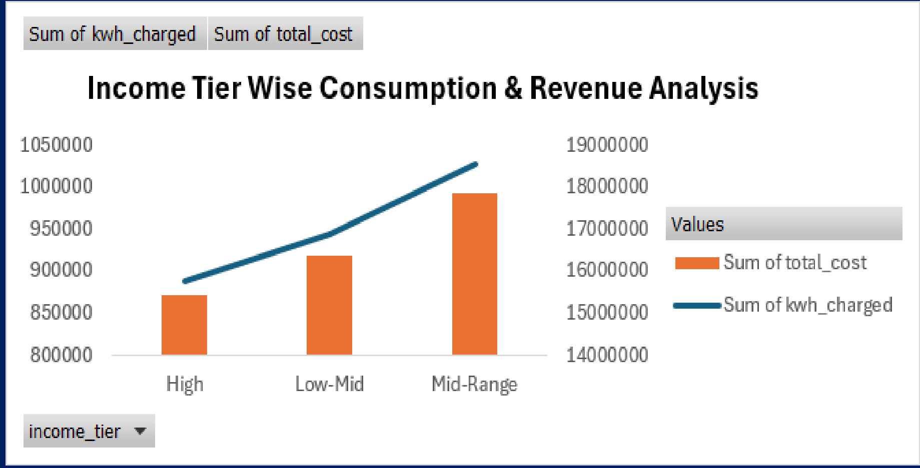
# Customer and Demand Behavior


**8,000**
Unique Customer Count

**14**
Sessions Per Customer

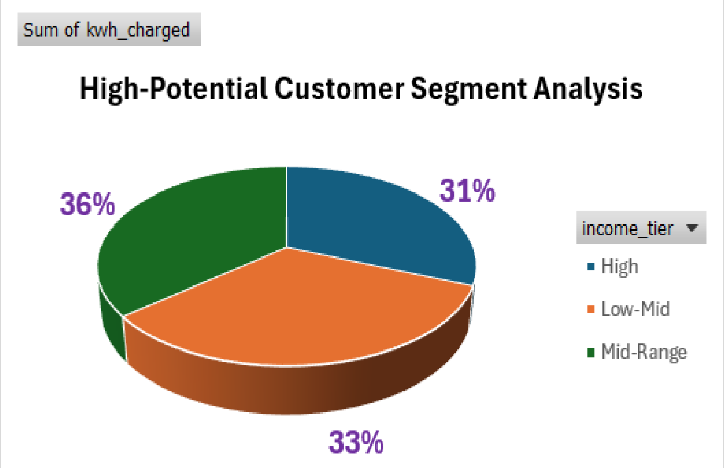
**357**
Average kwh per customer(Kwh)

**6,212**
Revenue per customer



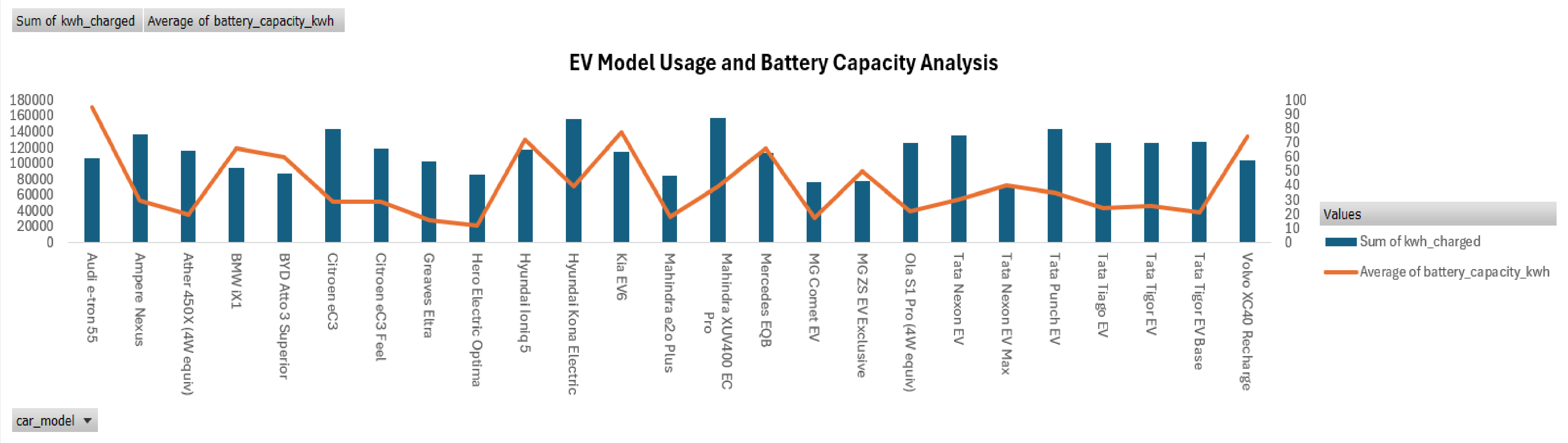

## Usage By Income Tier

| Income Tier | Sum of Kwh Charged |
|---|---|
| High | 888659.4183 |
| Low-Mid | 943602.867 |
| Mid-Range | 1027509.41 |




income_tier
- High
- Low-Mid
- Mid-Range

car_model
- Audi e-tron 55
- BMW iX1
- BYD Atto 3 Super...
- Citroen eC3
- Citroen eC3 Feel
- Greaves Eltra
- Hero Electric Opt...
- Hyundai Ioniq 5
- Hyundai Kona El...
- Kia EV6
- Mahindra e2o Plus
- Mahindra XUV40...
- Mercedes EQB
- MG Comet EV



# FINANCIAL PERFORMANCE AND EXPANSION STRATEGY (ROI FOCUS)

## PROBLEMS TO ADDRESS

- Which districts and projects deliver the highest ROI?
- What is the payback period for different charging setups?
- How do costs such as CAPEX, land lease, and infrastructure impact profitability?
- Which locations should be prioritized for new station deployment based on demand and ROI?
- What is the gap between existing infrastructure and projected EV demand?

## KPIs TO TRACK

- Total CAPEX per project
- Annual revenue and EBITDA
- Payback period (with and without subsidy)
- 5 year ROI percentage
- Existing stations vs projected EVs
- Demand supply gap (projected EVs vs projected plugs)
- Priority tier classification

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# FINANCIAL PERFORMANCE INSIGHTS

ROI DRIVERS & INVESTMENT RECOVERY ANALYSIS

## 1 WHICH DISTRICTS AND PROJECTS DELIVER THE HIGHEST ROI?

**KEY INSIGHTS**

- Projects like BLR-PRJ055, BLR-PRJ086 & BLR-PRJ041 deliver ROI above 180%.
- Bommanahalli, Marathahalli & Banashankari are the top ROI districts.
- Dasarahalli, Kengeri & Anekal show negative ROI due to low utilization and inefficiencies.

**BUSINESS INTERPRETATION**

- Commercial & IT hubs drive higher profitability through better utilization.
- Low-performing areas need operational optimization and demand-generation initiatives.

**BUSINESS CONCLUSION**

High-demand urban districts deliver the strongest ROI, while low-performing regions require utilization improvement to achieve profitability.

## 2 WHAT IS THE PAYBACK PERIOD FOR DIFFERENT CHARGING SETUPS?

**KEY INSIGHTS**

- AC Level-2 chargers have the fastest payback period and lower operational risk.
- DC Fast & Ultra-Fast chargers need longer recovery due to higher infrastructure costs.
- Government subsidies reduce payback periods across all charger types.

**BUSINESS INTERPRETATION**

- AC chargers are ideal for residential and workplace charging.
- DC Fast chargers are better for highways, commercial hubs & long-term strategic deployment.

**BUSINESS CONCLUSION**

AC Level-2 chargers offer the quickest investment recovery, while DC fast-charging infrastructure provides long-term strategic value despite higher capital costs.

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# FINANCIAL & DEPLOYMENT STRATEGY INSIGHTS

Cost Impact, Deployment Priorities & Infrastructure Gap

## 3 HOW DO COSTS IMPACT PROFITABILITY?

### KEY INSIGHTS

- DC Fast & Ultra-Fast chargers have higher CAPEX & infra costs → longer payback.
- AC Level-2 chargers deliver faster ROI due to lower setup & ops costs.
- High land lease in commercial hubs, but strong demand improves returns.
- Low-utilization areas (Anekal, Dasarahalli, Byatarayanapura) struggle to be profitable.

### BUSINESS INTERPRETATION

Profitability depends on balancing:

- CAPEX & infrastructure cost
- Land lease expenses
- Station utilization
- Charging demand

### BUSINESS CONCLUSION

Optimized cost + high utilization = strong profitability. High-cost, low-utilization locations need operational improvement.

## 4 WHICH LOCATIONS SHOULD BE PRIORITIZED FOR NEW DEPLOYMENT?

### HIGH PRIORITY LOCATIONS

HSR Layout | Jayanagar | Rajajinagar | Banashankari | Yelahanka

Strong EV demand + Positive ROI

### EMERGING GROWTH AREAS

Indiranagar | Koramangala | Whitefield

High future EV demand, need operational optimization

### BUSINESS INSIGHT

- High-density commercial zones
- Better utilization & faster payback
- Stronger long-term profitability

### BUSINESS CONCLUSION

Prioritize high-demand urban locations with strong ROI & projected EV growth.

## 5 WHAT IS THE GAP BETWEEN INFRASTRUCTURE & DEMAND?

CURRENT PLUGS **390** – FUTURE GAP **677** ADDITIONAL PLUGS REQUIRED = PROJECTED DEMAND **1,067**

### HIGH GAP LOCATIONS

Jayanagar | Rajajinagar | HSR Layout | Indiranagar | Koramangala

High EV growth but insufficient infrastructure

### LOW GAP LOCATIONS

Anekal | Byatarayanapura | Rajarajeshwari Nagar

Lower infrastructure pressure currently

### BUSINESS CONCLUSION

High-growth districts need major charging infrastructure expansion to meet future demand and avoid congestion.

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Lighting Up Lives!

Manage Costs Smartly | Deploy in the Right Locations | Bridge the Infrastructure Gap | Drive Higher ROI & Sustainable Growth

# KPIs MEASURES

| | | | |
|---|---|---|---|
| 1 | Total CAPEX per project | = | SUM('ROI Bengaluru Districts'[Total_CAPEX_Lakh]) / COUNT('ROI Bengaluru Districts'[Project_ID]) |
| 2 | Annual revenue | = | SUM('ROI Bengaluru Districts'[Annual_Revenue_Lakh]) |
| 3 | EBITDA | = | SUM('ROI Bengaluru Districts'[Annual_EBITDA_Lakh]) |
| 4 | Payback period | = | SUM('ROI Bengaluru Districts'[Payback_Period_Yrs]) |
| 5 | Payback period without subsidy | = | SUM('ROI Bengaluru Districts'[Adj_Payback_After_Subsidy]) |
| 6 | 5 year ROI percentage | = | SUM('ROI Bengaluru Districts'[5yr_ROI_Pct]) / COUNT('ROI Bengaluru Districts'[5yr_ROI_Pct]) |
| 7 | Existing stations vs projected EVs | = | SUM('ROI Bengaluru Districts'[Projected_EVs_District]) / SUM('ROI Bengaluru Districts'[Existing_Stations_District]) |
| 8 | Demand supply gap | = | SUM('ROI Bengaluru Districts'[Projected_EVs_District]) / SUM('ROI Bengaluru Districts'[Projected_Plugs_District]) |

# Financial Performance and Expansion Strategy (ROI Focus)

**142**
Total CAPEX per project (In Lakh)

**3,239**
EBITDA (In Lakh)

**4.61**
Sum of Payback Period

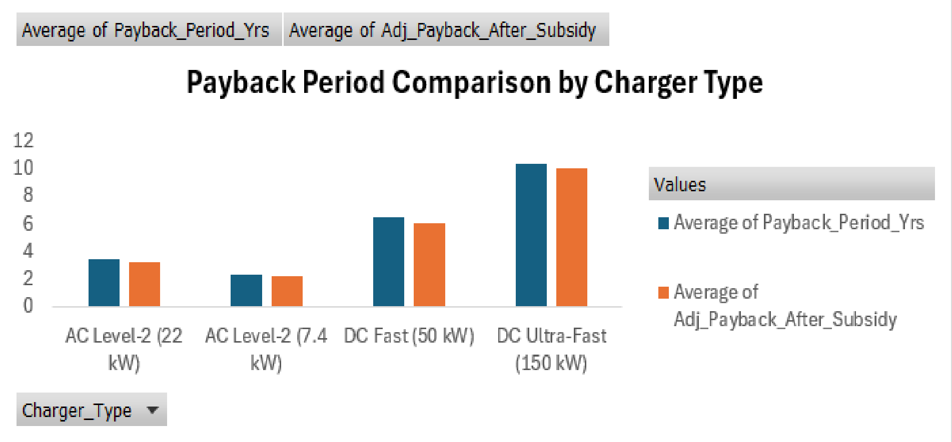
**4.34**
Sum of Payback Period After Subsidy

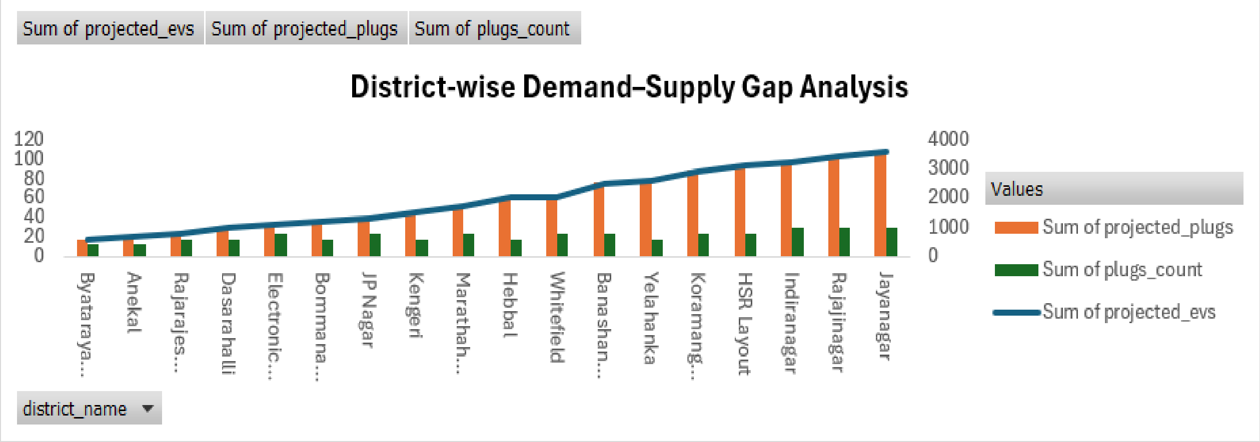
**41.43**
Avg. 5 year ROI Percentage

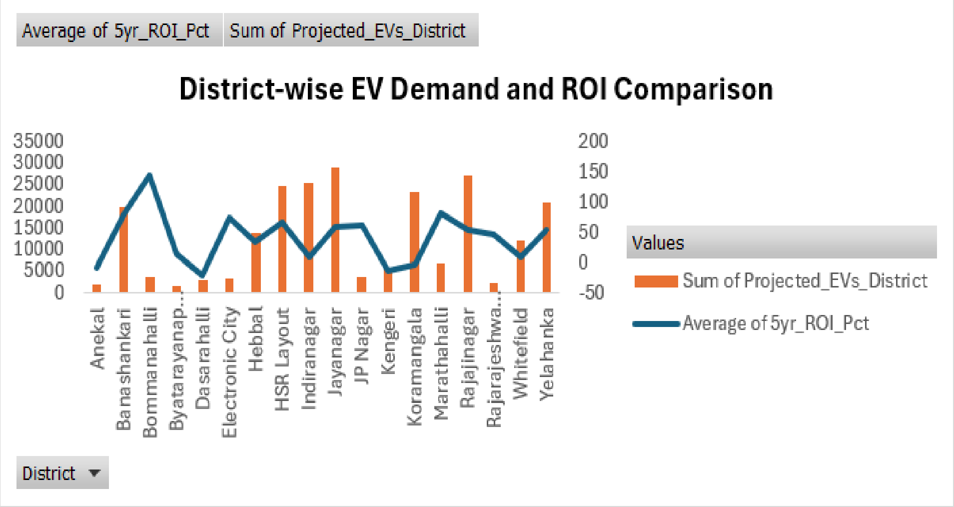
**4,629**
Annual Revenue (In Lakh)

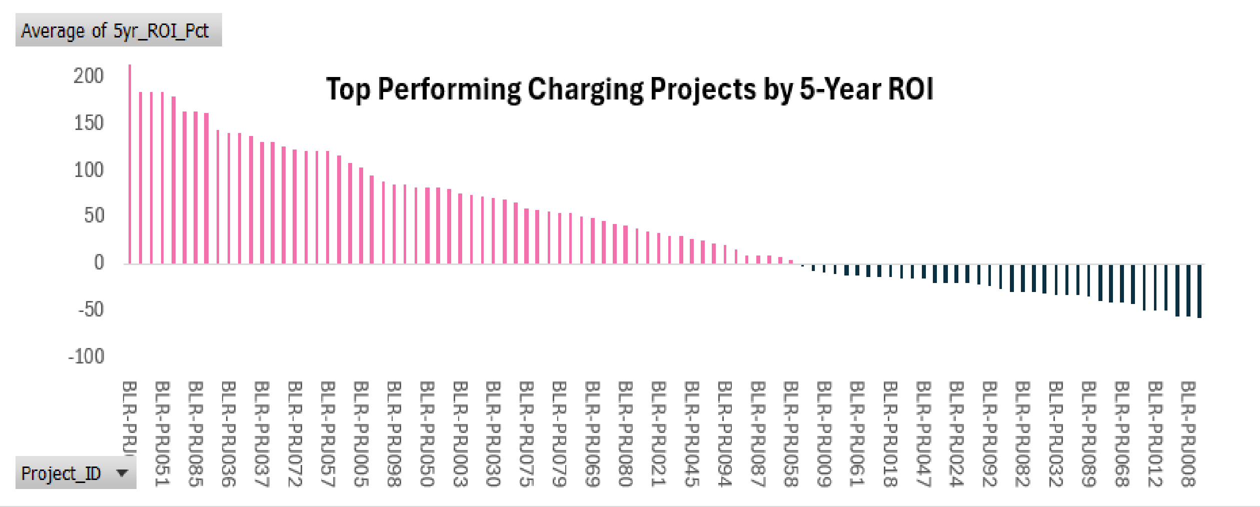
**33**
Demand Supply Gap

**610**
Existing stations vs Projected EVs




District
- Anekal
- Banashankari
- Bommanahalli
- Byatarayanap...
- Dasarahalli
- Electronic City
- Hebbal
- HSR Layout
- Indiranagar
- Jayanagar

Charger_Type
- AC Level-2 (22 kW)
- AC Level-2 (7.4 kW)
- DC Fast (50 kW)
- DC Ultra-Fast (15...

# FINAL BUSINESS CONCLUSION

Evaluating Performance, Profitability & Expansion Potential of EV Charging Infrastructure

### 1 Performance & ROI

- Top projects (BLR) achieve ROI >180%
- Highest ROI: Bommanahalli, Marathahalli, Banashankari
- Low ROI: Dasarahalli, Kengeri, Anekal

### 2 Payback by Charger Type

- AC Level-2: 2.2–3.2 years
- DC Fast: ~6.0 years
- DC Ultra-Fast: ~10 years
- Subsidies improve payback

### 3 Cost Impact on Profitability

- DC Fast & Ultra-Fast → Higher CAPEX, lower ROI
- AC Level-2 → Faster ROI
- High land lease in commercial hubs boosts revenue

### 4 Deployment Priorities

**High Priority:**

HSR Layout, Jayanagar, Rajajinagar, Banashankari

**Growth Areas:**

Indiranagar, Koramangala, Whitefield

### 5 Infrastructure Gap & Expansion Need

| Current Plugs | Future Gap | Projected Demand |
|---|---|---|
| 390 | 677 | 1,067 |

- Focus on high-demand urban zones
- Optimize low-utilization areas

## Overall Business Conclusion

Right mix of locations, charger types & investments will maximize profitability and support scalable EV growth.

**Invest Smartly** | **Operate Efficiently** | **Grow Sustainably**

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# FUTURE DATA AND LIMITATIONS

## FUTURE DATA REQUIREMENTS

- Real-time maintenance and downtime tracking data.
- Customer satisfaction and loyalty data.
- Dynamic pricing and electricity tariff information.
- Competitor infrastructure and market penetration data.
- Seasonal and weather impact analysis.

## CURRENT LIMITATIONS

- Limited historical data duration.
- Missing customer feedback metrics.
- No predictive forecasting model included.

# THANK YOU

## Powering a Sustainable Future with Smart EV Charging Solutions

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|---|---|---|---|

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Together, let's drive the change towards a cleaner, greener and smarter future.